Diagonal Power Certificates for Diophantine Consistency

A 214-variable integer quartic with coefficient height 219,813

Research prepared for Vladimir Reshetnikov's ProveIt project
AI-assisted mathematical research and reproducible arithmetic audit

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Abstract

We give an explicit polynomial $Q_{214} \in \mathbb{Z}[x_0, \ldots, x_{213}]$, of degree four and coefficient height 219,813, for which ZFC proves that absence of an integer zero is equivalent to its usual arithmetized consistency statement. The polynomial is the sum of squares of 153 displayed quadratic residuals; its collected expansion has 5,656 monomials. The construction modifies ProveIt package 173's direct C222 consistency encoding. Its growth certificate is replaced by

$$t = 464 + z^2, \qquad K = 464t^2, \qquad F = K^K.$$

The power is represented by the existing ordinary polynomial Pell certificate: the already authenticated index K also serves as the base. A positive-coefficient estimate for the actual C222 formula codes proves that this growth suffices, allowing one intermediate variable and one quadratic residual to be removed. At 214 variables and degree four, the new height is more than five million times smaller than that of package 173's existing 214-variable companion, at the cost of 170 additional monomials. Earlier self-power constructions are explicitly credited. This is a concrete tradeoff within a direct small-coefficient construction, not a claim of global simplicity or a new universal-polynomial record. The ordinary proof uses identified inherited Pell and logical calibration theorems. Exact independent arithmetic checks accompany it; a new end-to-end proof-assistant formalization is not claimed.

1 The equation and the precise result

Definition 1.1. For the 153 quadratic residuals printed in Appendix B, set

$$Q_{214}(x_0, \ldots, x_{213}) := \sum_{j=0}^{152} \mathcal{R}_j(x_0, \ldots, x_{213})^2 = 0, \qquad (x_0, \ldots, x_{213}) \in \mathbb{Z}^{214}. \tag{1}$$

Every variable is independently unrestricted over the integers. No positivity, divisibility, exponentiation, or omitted side condition is part of the definition.

The complete quadratic coefficient list is `direct/diagonal214/equation.json`; the complete collected quartic is `direct/diagonal214/expanded_polynomial.json`. The appendix is generated from the former and checked against it. Thus (1) specifies an ordinary fixed polynomial, rather than a polynomial schema or an exponential Diophantine equation. Powers with variable exponents below occur only in its mathematical interpretation.

Theorem 1.2. *Using the identified C222/Pell and C222/ZF/ZFC calibration theorems of ProveIt, the following equivalence is provable in ZFC:*

$$\left(\neg \exists \mathbf{x} \in \mathbb{Z}^{214}\ Q_{214}(\mathbf{x}) = 0\right) \quad \longleftrightarrow \quad \mathrm{Con}(\mathrm{ZFC}). \tag{2}$$

The polynomial has exact degree four, 5,656 nonzero collected monomials, and coefficient height $\mathrm{H}(Q_{214}) = 219{,}813$.

Here Con(ZFC) means that no finite derivation of falsity exists in the fixed effective first-order ZFC presentation used by ProveIt. This is the syntactic consistency predicate, not an assertion of external truth or existence of a transitive set model. The ordinary proof establishes the positive existence equivalence

$$\exists \mathbf{x} \in \mathbb{Z}^{214}\, Q_{214}(\mathbf{x}) = 0 \quad \longleftrightarrow \quad \mathrm{Ref}_{\mathrm{ZFC}}; \tag{3}$$

negation gives (2). No consistency assumption is required to prove these equivalences. In particular, this report does not prove that the displayed equation lacks solutions.

The meaning of "using" in the theorem is ordinary mathematical reliance on stated earlier theorems, not an extra consistency hypothesis. The new arithmetic changes are proved here. The inherited unbounded logical calibration is credited and its formalization boundary is stated in Section 8.

1.1 What improves, and what does not

The parent is the principal equation of package 173, `173-crt-index-anchor-215`, at repository commit `02a78a08603cbae248c19c23a75907c6c6bd6443` [1]. All equations in the following comparison have degree four and use unrestricted integer variables.

Equation	Variables	Residuals	Monomials	Coefficient height
Package 173, principal P_{215}	215	154	5,660	91,679
Package 173, companion P_{214}	214	153	5,486	1,099,511,627,852
This report, Q_{214}	**214**	**153**	**5,656**	**219,813**

Relative to P_{215}, one variable and one residual disappear, while height increases. Relative to the existing P_{214}, height falls by a factor greater than five million, while support increases by 170 monomials, about 3.10%. Package 173 also has a 213-variable companion, whose height is 2,417,851,639,242,452,488,945,685. The broader repository contains much smaller variable counts with other degree and coefficient tradeoffs. Accordingly, "simplest" has no established single optimum here. The defensible claim is the explicit measured improvement in the table.

The general diagonal-growth idea $F = t^t$ already occurs in package 170 and its sources [2]. The new contribution is the source-specific label majorant, the amplified diagonal $F = K^K$ with $K = 464t^2$, its application to the range-free C-anchor interface, and the fully emitted low-height endpoint. Neither the general self-power idea nor the inherited C-anchor construction is claimed as new.

2 A literal five-row change

The parent coordinate chart has been optimized by translations, so historical variable names are not by themselves semantic definitions. The coordinates that matter here are exactly

$$\begin{aligned} t &= x_0, & F &= x_2, & C &= 2x_{16}, & K &= x_{93}, \\ W &= x_{131}, & A &= x_{132}, & E &= x_{139}, & q_{\mathrm{pow}} &= x_{140}, \\ D_K &= x_{181}, & Y_K &= x_{182}, & X_K &= x_{183} + 1, & z &= x_{185}. \end{aligned}$$

A is the common signed Pell center, C the actual encoded row-index representative, and W the shared frame. In particular, x_{183} is $X_K - 1$, not X_K. The unchanged rows 126 and 127 say

$$1 - AY_K - D_K + x_{183} = 0, \tag{4}$$

$$-1 + 2D_K - D_K^2 + 2D_K x_{183} + Y_K^2 = 0. \tag{5}$$

Substituting $X_K = x_{183} + 1 = D_K + AY_K$ in (5) gives $X_K^2 - (A^2-1)Y_K^2 = 1$. The retained cap and orientation certificates authenticate its canonical index as K.

The old semantic growth data were

$$t = 128 + 4z^2, \qquad T_2 = t^2, \qquad K = T_2^2 = t^4, \qquad F = t^K.$$

The new data are

$$\boxed{t = 464 + z^2, \qquad K = 464t^2, \qquad F = K^K.} \tag{6}$$

To implement this change, retain coordinates $x_0, \ldots, x_{213}$, delete $x_{214} = T_2$, and change only the following parent rows:

Row	New quadratic residual
96	$1 + K^2 - 2KA + E$
97	$F - D_K - KY_K + Eq_{\mathrm{pow}}$
128	$\mathcal{R}_{128}^{\mathrm{old}} + t^2 - K^2$
130	$t - 464 - z^2$
152	$K - 464t^2$

Delete old row 153. Old rows 152 and 153 were $x_{214} - t^2$ and $K - x_{214}^2$. No other row used x_{214}; no other row uses the horizon root z. All other retained residuals are identical to the parent.

Rows 96 and 97 are the same singleton Pell power certificate with base changed from t to K. The index is still K, so no new principal pair is introduced. The actual square frame formerly contained the growth base t and the index K. It now contains K in both roles. The change $+t^2 - K^2$ in row 128 performs exactly this replacement; deleting the existing index occurrence would be incorrect. The independent reviewer reconstructed all 35 frame atoms, including affine translations and the two occurrences of K.

This transformation preserves existence of witnesses through their finite-refutation meaning. It is not a pointwise substitution between complete old and new zero sets: the horizons and huge growth values change. Both directions must therefore be proved semantically.

3 A certified bound for the actual formula labels

The source calculus is the same square-shell recoding of C222 used by the parent. Its constructor polynomials are

$$I(a,b) = 2\big((a+b)^2 + a\big), \qquad M(a,b) = I(a,b) + 1,$$
$$A_{\mathrm{syn}}(a,b) = 2\big((a+b)^2 + 3a + 4b + 3\big), \qquad E_{\mathrm{syn}}(a,b) = A_{\mathrm{syn}}(a,b) + 1.$$

The syntax constructor A_{syn} is distinct from the Pell center A. Let $J(a,b,c,d)$ be the sum of the nine unchanged axiom-code expressions and the five expressions $a, b, I(a,b), A_{\mathrm{syn}}(a,b), 1$. Every possible nonzero local conclusion or premise label is bounded by J. Write $j(s) = J(s,s,s,s)$. This is a mathematical majorant only: no new equation variable or range request for J is added.

3.1 The finite arithmetic certificate

The authenticated source `majorant/shell_source.py` [3] is 2,955 bytes, with SHA-256 recorded in Appendix A. Its actual `codes_only` path is executed through a positive arithmetic DAG by `majorant/check_label_majorant.py`. The DAG has 312 nodes. All coefficients are nonnegative, and the exact degrees of its fourteen summands, in source order, are

$$16, 8, 8, 8, 8, 32, 16, 8, 524288, 1, 1, 2, 2, 0.$$

Thus $\deg j = 524288$. The decisive exact inequalities are

$$\operatorname{bitlen}\big(j(464)\big) = 5{,}692{,}913, \tag{7}$$
$$\operatorname{bitlen}\big(464^{645888}\big) = 5{,}721{,}264. \tag{8}$$

In particular,

$$j(464) < 2^{5692913} < 2^{5721263} \le 464^{645888}. \tag{9}$$

The finite certificate uses exact integer arithmetic with outward-rounded dyadic bounds, not floating-point logarithms. For example, it supplies

$$j(464) \le 12187191352475505376 \cdot 2^{5692849},$$
$$464^{645888} \ge 14263689019699129227 \cdot 2^{5721200}.$$

The matching lower and upper bounds locate each integer in a unique binary interval. The optional `--exact` check evaluates the full integers and confirms the bit lengths; that check was also run for this package. It is unnecessary to expand the degree-524288 polynomial.

Lemma 3.1 (Positive-coefficient extrapolation)**.** *Let $p(X) = \sum_{i=0}^{D} a_i X^i$ with $a_i \ge 0$. If $t \ge c > 0$, $d \ge D$, and $p(c) < c^d$, then $p(t) < t^d$.*

Proof. For each $i \le D$, $t^i = c^i(t/c)^i \le c^i(t/c)^D$. Hence

$$p(t) \le p(c)(t/c)^D < c^d(t/c)^D \le t^d.$$

Clearing denominators gives an equivalent integer argument. □

Set

$$c_* = 464, \qquad d_* = 3c_*^2 = 645888, \qquad L(t) = t^{d_*}. \tag{10}$$

Since $d_* \ge 524288$, Lemma 3.1 and (9) give $j(t) < L(t)$ for every integer $t \ge 464$. The retained independent parameter ranges give $0 \le a, b, c, d < t$ at every decoded local row. Monotonicity therefore yields

$$J(a, b, c, d) \le j(t-1) < j(t) < L(t). \tag{11}$$

This is a uniform proof for all possible witnesses. The single finite evaluation is a certificate for the hypothesis of an unbounded positive-coefficient lemma, not evidence from sampling.

The same certificate finds that $c = 462$ fails this particular sufficient test, while $c = 463$ succeeds. We choose the convenient even coefficient 464, which supplies the parity margin used below. This observation is not a minimality proof for the chosen constant or for any Diophantine construction.

4 The amplified diagonal growth lemma

Lemma 4.1. *If $t \ge c \ge 3$ are integers, then $c^t \ge t^c$.*

Proof. At $t = c$ there is equality. For every $n \ge c$,

$$(1 + 1/n)^c \le (1 + 1/c)^c \le \sum_{r=0}^{c} \frac{1}{r!} < 3 \le c.$$

The strict bound follows from $r! \ge 2^{r-1}$ for $r \ge 2$ and the finite geometric sum. Thus $(n+1)^c < cn^c$. Multiplying the induction hypothesis $c^n \ge n^c$ by c proves $c^{n+1} \ge (n+1)^c$. □

Proposition 4.2. *The semantic data* (6) *satisfy*

$$F \geq L(t)^t = t^{645888t}. \tag{12}$$

Proof. Put $c = c_* = 464$. Since $t \geq c$ and $K = ct^2$,

$$\begin{aligned} F = K^K &= (ct^2)^{ct^2} = (c^t)^{ct}\, t^{2ct^2} \\ &\geq t^{c^2t+2ct^2} \geq t^{3c^2t} = L(t)^t. \end{aligned}$$

The first inequality uses Lemma 4.1; the second uses $t \geq c$. □

The useful design principle is that a very large authenticated index can also be its own base. The parent already paid for the K-index Pell pair. Increasing its numerical role, while bounding the actual source labels more accurately, replaces two quadratic growth-index gates by one. This changes the growth certificate without changing the logical calculus or introducing a machine simulation.

5 Why the growth bound is enough

The following sharp lemma is inherited from package 173. Its proof is included because it explains exactly what the new certificate must supply.

Lemma 5.1 (Finite-product separation)**.** *Let t, L, F be positive integers, $L \geq t$ and $F \geq L^t$. Let U, V be finite multisets in $\{1, \ldots, L\}$, with $|U| \leq t$ and $|V| \leq 2t + 1$. If*

$$\prod_{u \in U} (1 + uF) = \prod_{v \in V} (1 + vF),$$

then $U = V$ with multiplicities.

Proof. First recover the factor counts. If they differ, set $r = \min(|U|, |V|) \leq t$. The larger product is at least $(F + 1)^{r+1}$; the smaller is at most $(1 + LF)^r \leq L^r(F + 1)^r$. Equality would give $F + 1 \leq L^r \leq F$, a contradiction. Thus $|U| = |V| = m \leq t$. The case $m = 0$ is immediate.

Replace F by an indeterminate Z. Every coefficient of either product, in each degree $0 \leq k \leq m$, is a positive integer at most

$$\binom{m}{k} L^k \leq m^{m-k} L^k \leq L^m \leq F.$$

Here $\binom{m}{k} = \binom{m}{m-k} \leq m^{m-k}$. Differences of corresponding coefficients have absolute value strictly less than F, including when one original coefficient equals F.

If the difference polynomial is nonzero, let r be its least nonzero degree, with coefficient b_r. Equality at $Z = F$, divided by F^r and reduced modulo F, forces $F \mid b_r$, contradicting $0 < |b_r| < F$. The polynomials are identical. Unique factorization in $\mathbb{Q}[Z]$, or the distinct rational roots $-1/j$, recovers every label multiplicity. □

The decoded local rules have conclusion v_i and two premises p_i, q_i; missing premises are padded by truth, whose code is zero. Put $w_j = 1 + Fj$, so $w_0 = 1$. The retained pending-state update is

$$S_{i+1} w_{v_i} = S_i w_{p_i} w_{q_i}.$$

The beta endpoints give the initial state $F + 1$ and final state 1. The possible boundary case $y = F$ would make the initial residue zero; the positive weights would propagate zero to the final

state, a contradiction. Hence $y > F$, all states are positive, and multiplication and cancellation of the t updates gives

$$\prod_{i<t} w_{v_i} = (1+F) \prod_{i<t} w_{p_i} w_{q_i}. \tag{13}$$

After discarding truth factors, the left side has at most t factors and the right side at most $2t+1$. Equations (11) and (12) allow Lemma 5.1 to be applied. Consequently conclusion multiplicity minus premise multiplicity is one at falsity label 1 and zero at every other positive label.

Take the closure of truth under the finitely many instantiated rules. If falsity were outside it, each rule with an outside conclusion would have an outside premise. Its contribution to the total outside-label balance would be nonpositive; the total could not be one. This contradicts the recovered multiplicities. The argument extracts a finite C222 refutation even if the selected rule multiset contains cycles or repeated premises. No topological ordering is assumed.

6 Soundness: every integer zero gives a refutation

Assume $Q_{214}(\mathbf{x}) = 0$ for an arbitrary integer tuple. Since it is a sum of integer squares, each residual vanishes. Row 130 gives $t \geq 464$, and row 152 gives $K = 464t^2 > 0$, with K even. The required order of the remaining deductions is important.

6.1 Recover the canonical powers before postponed signs

The unchanged frame rows give a large nondegenerate signed center A; the frame contains every actual requested base, index, and output. The parent's C-anchor theorem [1] classifies the anchor pair and bounds all principal indices before the sign of the actual CRT representative C is known. Its squared cap and bounded-error orientation then authenticate every principal pair, including the retained K pair. This theorem remains applicable: the modified frame has the same form and still contains all actual atoms. In particular, both the growth base and growth index are the already positive K.

Rows 96 and 97 are exactly the singleton power certificate for this base and index. They therefore force $F = K^K$. Thus $F > 0$ is even before the original sign bootstrap is used. Since $C = 2x_{16}$ is even and the retained divisibility is $B \mid C + F + 1$, B is nonzero and odd. The self-sized radix multiplier remains 2 modulo 4, so its first nontrivial power has a base of absolute value at least two. The unchanged singleton and pooled power arguments can now recover the other powers and their signs in the original order. None of this assumes the desired semantic power equality before applying its certificate.

6.2 The unchanged range and local-row interface

All nine independent local-column range requests remain: four formula parameters, two pending states, two beta quotients, and the encoded row index. The only derived-label range previously removed by package 173 is still absent. Section 3 supplies its needed upper bound. The local Horn polynomial, both beta endpoint congruences, the binomial range certificates, and every-prime CRT compatibility are unchanged.

The inherited sign and beta bounds give $y \geq F$ and $x \geq 2 + (t+1)y$, where x denotes the beta code (not a free new coordinate). For the local polynomial let $R_0 = 2 + t + x + y + F$. The parent's uniform bound is

$$0 \leq \text{Local} < R_0^{2^{23}}.$$

The new growth is stronger than $F \geq t^t$, and $t \geq 464 > 16$; hence $x > 2^{32}$ and $R_0 < 2x$. If $2^k \leq x < 2^{k+1}$ with $k \geq 32$, then $2^{23}(k+2) < 2^k \leq x$, so Local $< 2^x$. Thus the unchanged prime-

power selector still upgrades local congruences to exact zero rows. This is the same unbounded local estimate, not a check at particular witness values.

For the binomial-selector radius, put $\Lambda_t = \operatorname{lcm}(1, \dots, t)$. The uniform estimate

$$\Lambda_t^2 \le (t!)^2 \le t^{2t} \le F \tag{14}$$

follows from Proposition 4.2. It avoids any parity restriction on t. An earlier, sharper estimate proved only for even horizons is not being used.

Consequently the decoded local rows satisfy the update and endpoints used in Section 5. Product separation and the finite Horn-closure argument yield a C222 refutation. We have proved

$$\exists \mathbf{x} \in \mathbb{Z}^{214}\, Q_{214}(\mathbf{x}) = 0 \quad \Longrightarrow \quad \mathrm{Ref}_{\mathrm{C222}}\,. \tag{15}$$

7 Completeness: every finite refutation gives a zero

Start with a finite square-shell C222 refutation. Choose $z \in \mathbb{N}$ so large that $t = 464 + z^2$ exceeds its local length and every independent source parameter. These horizons are cofinal. Pad with legitimate truth steps as in the parent, and set $K = 464t^2$, $F = K^K$. All source labels and all finite pending-multiset states are now fixed.

7.1 The parity change is harmless

Although t need not be even, $464 = 16 \cdot 29$ gives

$$v_2(K) = 4 + 2v_2(t), \qquad v_2(F) = Kv_2(K) \ge 4K > 2t. \tag{16}$$

In particular F contains more than the entire power of two dividing $t!$. Let O be the odd part of $t!$. Choose a sufficiently large positive odd multiple u of O and set $y = Fu$. Then $t! \mid y$, $y > F$, and y exceeds all genuine pending states. It also has the required literal shape $F(2w + 1)$.

The beta moduli $1 + (i + 1)y$ are pairwise coprime. Indeed a common divisor of the ith and jth moduli divides $(j - i)y$ and is coprime to y; it therefore divides $j - i$, which divides $t!$ and hence y. The divisor is one. Ordinary finite CRT supplies the beta code, chosen cofinally large in its class.

Now set $B = \binom{F+t}{t}$ and $N = B^x$. For each $1 \le i \le t$, (16) implies $v_2(F + i) = v_2(i)$. The binomial product formula gives $v_2(B) = 0$, so B, N, and N^2 are odd. The parity-free radius estimate (14) also holds. Thus both directions of the inherited generalized binomial CRT theorem apply: every row has a selector, and every prime dividing B has a compatible row at its required full prime-power precision [4].

7.2 Choose the reused anchor in a noncircular order

The local-column CRT conditions determine nine residue classes modulo N^2, one for each independent column. Choose the eight ordinary representatives first. Then choose the actual row-index representative C in its prescribed class, positive and even, greater than all other requested nominal indices, including K. This is always possible: N^2 is odd, so the permitted even values form a cofinal progression of step $2N^2$.

Only after this choice define the self-sized radix multiplier, its power outputs, the actual square frame W, and its positive canonical Pell center A. The modulus N^2 was fixed before choosing C and does not depend on this later frame. Increasing C preserves all local CRT conditions and strengthens the self-sized radix bounds. This is the cofinal-C argument of the parent, with the new index K included in the finite list to be dominated.

The parent's cap and orientation completeness proof now supplies every auxiliary Pell witness. The even index C provides the same odd auxiliary quotient; its canonical ordinate dominates the retained finite sum of the other ordinates. Nonnegative slacks retain their ternary representations. The new first power request uses canonical base K and index K, and output $F = K^K$, so its divisor quotient is integral. All other singleton, pooled, range, and local witnesses are unchanged in form. Set the new index wire to $464t^2$ and omit the old private t^2 wire. This constructs a zero of every new residual. Therefore

$$\mathrm{Ref}_{\mathrm{C222}} \quad \Longrightarrow \quad \exists \mathbf{x} \in \mathbb{Z}^{214}\, Q_{214}(\mathbf{x}) = 0. \tag{17}$$

8 Logical calibration and the proof boundary

Equations (15) and (17) prove the arithmetic existence equivalence with the unchanged square-shell C222 calculus. The remaining logical chain is

$$\begin{aligned} \text{square-shell C222 refutation} &\iff \text{original C222 derivation of code 1} \\ &\iff \text{ZF refutation} \\ &\iff \text{ZFC refutation.} \end{aligned}$$

The first arrow is the parent's syntax recoding, a bijection fixing truth 0 and falsity 1 and transporting guarded rule instances. It is not a claim that square-shell arithmetic constructors literally equal the original numerical constructors.

The two named logical endpoints are

```
SetTheory.BusyBeaver.ZFC.KEndpoint.c222_iff_zf_refutation
SetTheory.BusyBeaver.ZFC.ConstructibleChoice.zf_refutation_iff_zfc_refutation.
```

They are pinned at commit `953fb4d82d4229f4efc5f9db4cc2bf38ae329bb4` [5, 6]. Copies of their source files are included as dependency excerpts, with byte and Git-blob hashes. Their transitive import closure is not bundled or newly compiled. The endpoint source file also contains machine theorems, but the named C222/refutation endpoint has no machine argument. This construction introduces neither a K215 computation nor a specialization of Jones' universal polynomial.

The new proof uses finite syntax, induction, ordinary integer inequalities, polynomial identities, finite CRT, and the identified Pell theorems. These are formalizable in ZFC and compose with the inherited calibration to prove (3) and Theorem 1.2. All reasoning is uniform over the internally coded integers and finite derivations of ZFC. It is not restricted to externally standard integers or proofs. No appeal to omega-consistency or external well-foundedness of a model is made.

The largest logical dependency not independently rederived here is the full C222/ZF calibration, including the seed-to-model lower direction. Its internally well-founded model construction must not be replaced by an assumption that consistency yields an externally transitive model. On the arithmetic side, the delicate inherited dependencies are signed Pell classification, squared divisibility, abscissa comparison, bounded orientation, and pooled-power error estimates. The retained parent note `inherited/parent173/ORDINARY_PROOF_DEPENDENCY_BOUNDARY.md` identifies their precise sources.

The result's status is therefore an ordinary mathematical proof with explicit inherited dependencies and exact finite arithmetic certificates. It is not a new end-to-end Lean or Rocq proof object. The supplied checks establish literal finite statements about the emitted equation and source majorant; they do not on their own prove the unbounded Pell, CRT, or logical interpretation.

9 Verification, scope, and further research

The producer authenticates the parent bytes before any transformation. Two separately implemented finite checkers import none of that producer's polynomial code. Between them they verify every changed row, every unchanged row, the complete shared frame, index-pair roles, cap and orientation terms, coordinate liveness, every coefficient of the collected quartic, and the exact source-bound label calculation. The root checker also checks all printed residuals, regeneration of the TeX appendix, deterministic reproduction of all six equation artifacts, and rejection of corrupted versions of the five risk-bearing rows. The standard-library checks passed both normally and with Python assertions disabled. Exact huge-integer evaluation was also performed.

The degree is at most four because every residual is quadratic. It is exactly four because x_{185} occurs only in $\mathcal{R}_{130} = x_0 - 464 - x_{185}^2$; the coefficient of x_{185}^4 in Q_{214} is one. All 214 variables occur in the collected polynomial. The residual list contains 1,040 nonzero terms and has coefficient height 464. Collecting all squares, including cancellations of mixed terms, gives the stated 5,656 monomials and height 219,813.

The useful next research questions are narrower than an unsupported claim of global simplicity. Can another already authenticated index also absorb a power base without increasing the shared frame cost? Can the source label majorant be made cheaper by changing the coding while preserving a fully identified logical calibration? Can a different Horn aggregation reduce the number of independent CRT columns? Each would require a complete soundness and completeness argument; finite numerical reductions alone do not establish it.

References

[1] V. Reshetnikov, *ProveIt, package 173: CRT-index anchor, 215/214/213 integer quartics.* Pinned repository commit `02a78a08603cbae248c19c23a75907c6c6bd6443`. Repository package and ordinary proof notes.

[2] V. Reshetnikov, *ProveIt, package 170: quotient carriers, diagonal-pool source proof.* Same repository snapshot. Earlier self-power construction.

[3] V. Reshetnikov, *ProveIt, package 112 intake: square-shell C222 source*, `ConZFC_Shell_Reciprocal_660/code/shell_source.py`, commit `953fb4d82d4229f4efc5f9db4cc2bf38ae329bb4`. Pinned source file.

[4] V. Reshetnikov, *ProveIt, package 151: factorial-free binomial CRT*, `report/main.tex`, same pinned logical/source commit. Generalized selector theorem and every-prime completeness.

[5] V. Reshetnikov, *ProveIt, KEndpoint.lean*, commit `953fb4d82d4229f4efc5f9db4cc2bf38ae329bb4`. C222/ZF refutation endpoint.

[6] V. Reshetnikov, *ProveIt, ConstructibleChoice.lean*, same commit. ZF/ZFC refutation transport.

A Reproduction and artifact identities

Run the following from the extracted package directory. The Python programs require only the standard library.

```
python3 -B verify.py --rebuild
python3 -B -O verify.py --rebuild --exact-majorant
python3 -B verification/review_diagonal.py
pdflatex -interaction=nonstopmode -halt-on-error report.tex
pdflatex -interaction=nonstopmode -halt-on-error report.tex
```

The first two commands check equation data and regenerate its printable definition; they do not compile a proof-assistant development. The last two compile this PDF using standard LaTeX packages. The evaluator `evaluate.py` accepts an explicit JSON list of 214 integer coordinates and evaluates either the quartic or all residuals, with an optional modulus; it is not a solution search or a consistency test.

Artifact	Role
`report.tex`, `residuals.tex`	Complete paper and all literal residuals
`direct/diagonal214/`	Exact equation, full expansion, metrics, variable map, source delta
`direct/build_diagonal.py`	Deterministic producer from authenticated parent
`direct/principal/`	Archived principal parent equation
`majorant/`	Authenticated source, positive-DAG checker, certificate, proof
`verify.py`	Independent artifact and delta verification
`verification/`	Separate review, checker, and execution receipts
`inherited/`	Credited proof dependencies and logical endpoint excerpts
`SOURCES.md`	Source paths, commits, and provenance

The exact SHA-256 identities are listed below. Line breaks in the display are typographical only; concatenate each pair of half-lines.

```
Parent equation:
8d919a1e5d6a802765b6d010c0140fb1d
83c3d36d211b209b7a8d2fb3f3fb6f6

New equation:
7ffda8ff6f609918dc3d1824a28ff718
8ec8c3ea2be83abb1e4bc5d465fd7894

Collected expansion:
171c6c26f97c0c0846f2aa4c6a8e134d
b6d4d2d459874b1e3b224bce67a8bcc4

Square-shell source:
59c68a354fe5cdbc04bde33720080db2
436cb51ae25e3485ac5ae45ff4230199
```

The equation digest authenticates the complete quadratic coefficient list, rather than a shortened display. Verification receipts report these same identities.

B Complete quadratic residuals

The following are all 153 residuals, indexed from 0 to 152. Every symbol x_i is one of the independently quantified integers $x_0, \ldots, x_{213}$. There are no external constants to instantiate and no omitted rows. Equation (1) is the sum of their squares. Rows that wrap continue on the next aligned line.

$$
\begin{aligned}
\mathcal{R}_{0} &= x_{0} - x_{0}x_{4} - x_{4} + x_{7} \\
\mathcal{R}_{1} &= -x_{2} - x_{4}x_{5} + x_{6}x_{7} \\
\mathcal{R}_{2} &= -1 + x_{1} - x_{6}x_{7} \\
\mathcal{R}_{3} &= -2x_{2}x_{213} - x_{3}x_{18} + x_{4} + 2x_{16} \\
\mathcal{R}_{4} &= -2 - x_{8} - x_{8}^{2} - 2x_{8}x_{9} - 2x_{9} - x_{9}^{2} + x_{19} \\
\mathcal{R}_{5} &= -2 - x_{8} - x_{8}^{2} - 8x_{8}x_{19} - 4x_{8}x_{20} - 8x_{19} - 16x_{19}^{2} - 16x_{19}x_{20} - 4x_{20} \\
&\quad - 4x_{20}^{2} + x_{21} \\
\mathcal{R}_{6} &= 1 + 2x_{21} - 4x_{21}^{2} + x_{22} \\
\mathcal{R}_{7} &= -2 - x_{9}^{2} - 2x_{9}x_{10} - x_{10}^{2} + x_{23} \\
\mathcal{R}_{8} &= -1 + 2x_{8} - x_{8}^{2} - 2x_{8}x_{10} + 2x_{10} - x_{10}^{2} + x_{24} \\
\mathcal{R}_{9} &= -7 + 33x_{8} - 35x_{8}^{2} - 96x_{8}x_{10} - 24x_{8}x_{19} - 4x_{8}x_{23} - 24x_{8}x_{24} + 16x_{9} \\
&\quad - 2x_{9}^{2} - 8x_{9}x_{23} + 32x_{10} - 64x_{10}^{2} - 32x_{10}x_{19} - 32x_{10}x_{24} + 4x_{19} \\
&\quad - 4x_{19}^{2} - 8x_{19}x_{24} + 12x_{23} - 4x_{23}^{2} + 8x_{24} - 4x_{24}^{2} + x_{25} \\
\mathcal{R}_{10} &= -16 + 14x_{8} - 2x_{8}^{2} - 8x_{8}x_{9} - 8x_{8}x_{23} + 8x_{9} + 8x_{23} - 4x_{25}^{2} + x_{26} - 2x_{43} \\
\mathcal{R}_{11} &= -6 + 16x_{10} - 16x_{10}^{2} - 16x_{10}x_{23} - 16x_{10}x_{24} + 8x_{23} - 4x_{23}^{2} - 8x_{23}x_{24} \\
&\quad + 8x_{24} - 4x_{24}^{2} + x_{27} \\
\mathcal{R}_{12} &= -5 + 24x_{10} - 16x_{10}^{2} - 16x_{10}x_{19} - 32x_{10}x_{24} - 16x_{10}x_{27} + 10x_{19} - 4x_{19}^{2} \\
&\quad - 16x_{19}x_{24} - 8x_{19}x_{27} + 24x_{24} - 16x_{24}^{2} - 16x_{24}x_{27} + 12x_{27} - 4x_{27}^{2} \\
&\quad + x_{28} \\
\mathcal{R}_{13} &= -12 - 29x_{8} - 37x_{8}^{2} - 24x_{8}x_{9} - 112x_{8}x_{10} - 4x_{8}x_{23} - 24x_{8}x_{24} - 39x_{9} \\
&\quad - 37x_{9}^{2} - 112x_{9}x_{10} - 24x_{9}x_{23} - 4x_{9}x_{24} - 80x_{10} - 128x_{10}^{2} - 32x_{10}x_{23} \\
&\quad - 32x_{10}x_{24} - 12x_{23} - 4x_{23}^{2} - 8x_{24} - 4x_{24}^{2} + x_{29} \\
\mathcal{R}_{14} &= -18 - 10x_{8} - 2x_{8}^{2} - 24x_{8}x_{9} - 32x_{8}x_{10} - 8x_{8}x_{23} - 72x_{9} - 72x_{9}^{2} \\
&\quad - 192x_{9}x_{10} - 48x_{9}x_{23} - 96x_{10} - 128x_{10}^{2} - 64x_{10}x_{23} - 24x_{23} - 8x_{23}^{2} \\
&\quad - 4x_{29}^{2} + x_{30} \\
\mathcal{R}_{15} &= -32 - 42x_{8} + 8x_{8}x_{10} + 16x_{8}x_{23} - 84x_{9} + 16x_{9}x_{10} + 32x_{9}x_{23} + 21x_{10} \\
&\quad - 3x_{10}^{2} - 4x_{10}x_{19} - 8x_{10}x_{23} + 21x_{19} - 8x_{19}x_{23} + 24x_{23} - 4x_{23}^{2} + x_{31} \\
&\quad - x_{59} \\
\mathcal{R}_{16} &= -1 - 24x_{8} - 64x_{8}^{2} - 160x_{8}x_{10} - 64x_{8}x_{23} - 32x_{8}x_{24} - 32x_{8}x_{31} - 26x_{10} \\
&\quad - 100x_{10}^{2} - 80x_{10}x_{23} - 40x_{10}x_{24} - 40x_{10}x_{31} - 8x_{23} - 16x_{23}^{2} \\
&\quad - 16x_{23}x_{24} - 16x_{23}x_{31} - 6x_{24} - 4x_{24}^{2} - 8x_{24}x_{31} - 4x_{31} - 4x_{31}^{2} + x_{32} \\
\mathcal{R}_{17} &= 1 - 4x_{19}^{2} + x_{20} \\
\mathcal{R}_{18} &= -2 - x_{8} - x_{8}^{2} + 8x_{8}x_{19} - 4x_{8}x_{20} + 8x_{19} - 16x_{19}^{2} + 16x_{19}x_{20} - 4x_{20} \\
&\quad - 4x_{20}^{2} + x_{33} \\
\mathcal{R}_{19} &= 1 + 2x_{33} - 4x_{33}^{2} + x_{34} \\
\mathcal{R}_{20} &= -x_{9}^{2} - 4x_{9}x_{34} - 4x_{34}^{2} + x_{35} \\
\mathcal{R}_{21} &= -8 - 20x_{8} - 16x_{8}^{2} - 64x_{8}x_{9} + 32x_{8}x_{19} - 40x_{9} - 64x_{9}^{2} + 64x_{9}x_{19} + 18x_{19} \\
&\quad - 16x_{19}^{2} + x_{36} \\
\mathcal{R}_{22} &= 8x_{9} - 62x_{9}^{2} - 92x_{9}x_{10} - 32x_{9}x_{23} - 32x_{9}x_{36} + 6x_{10} - 34x_{10}^{2} \\
&\quad - 24x_{10}x_{23} - 24x_{10}x_{36} - 4x_{23}^{2} - 8x_{23}x_{36} + 4x_{36} - 4x_{36}^{2} + x_{37}
\end{aligned}
$$

$$\mathcal{R}_{23} = 8x_8 - 64x_8^2 - 96x_8x_{10} - 32x_8x_{24} - 32x_8x_{37} + 6x_{10} - 36x_{10}^2 - 24x_{10}x_{24} - 24x_{10}x_{37} + 2x_{24} - 4x_{24}^2 - 8x_{24}x_{37} + 4x_{37} - 4x_{37}^2 + x_{38}$$

$$\mathcal{R}_{24} = -x_{10}^2 - 2x_{10}x_{11} - x_{11}^2 + x_{39}$$

$$\mathcal{R}_{25} = -32 + 22x_8 - 3x_8^2 - 8x_8x_9 - 8x_8x_{23} - 8x_8x_{24} - 4x_8x_{39} + 16x_9 - 8x_9x_{24} - 2x_{10}^2 - 8x_{10}x_{39} + 16x_{23} - 8x_{23}x_{24} + 22x_{24} - 4x_{24}^2 + 4x_{39} - 4x_{39}^2 + x_{40} - x_{43}$$

$$\mathcal{R}_{26} = 6 - 16x_8^2 - 36x_8x_{10} - 32x_8x_{24} - 40x_8x_{39} - 16x_8x_{40} - 22x_{10}^2 - 32x_{10}x_{24} - 48x_{10}x_{39} - 16x_{10}x_{40} - 2x_{24} - 16x_{24}^2 - 32x_{24}x_{39} - 16x_{24}x_{40} + 4x_{39} - 24x_{39}^2 - 16x_{39}x_{40} - 4x_{40}^2 + x_{41}$$

$$\mathcal{R}_{27} = x_{10} - 25x_{10}^2 - 40x_{10}x_{24} - 20x_{10}x_{27} - 16x_{24}^2 - 16x_{24}x_{27} - 4x_{27}^2 + x_{42}$$

$$\mathcal{R}_{28} = -1 + 8x_9 - 4x_9^2 - 8x_9x_{23} + 8x_{23} - 4x_{23}^2 + x_{43}$$

$$\mathcal{R}_{29} = -1 - 16x_9^2 - 8x_9x_{10} - 32x_9x_{23} - 16x_9x_{43} - x_{10}^2 - 8x_{10}x_{23} - 4x_{10}x_{43} - 16x_{23}^2 - 16x_{23}x_{43} - 4x_{43}^2 + x_{44}$$

$$\mathcal{R}_{30} = 3 + x_9 - 2x_9x_{10} - 4x_9x_{44} - 3x_{10}^2 - 8x_{10}x_{44} - x_{23} - 4x_{44}^2 + x_{45}$$

$$\mathcal{R}_{31} = -1 - 4x_{45} - 4x_{45}^2 + x_{46}$$

$$\mathcal{R}_{32} = -1 - x_{10}^2 - 8x_{10}x_{45} - 4x_{10}x_{46} - 16x_{45}^2 - 16x_{45}x_{46} - 4x_{46}^2 + x_{47}$$

$$\mathcal{R}_{33} = 2 + x_9 - x_9^2 - 4x_9x_{10} - 4x_9x_{47} - 4x_{10}^2 - 8x_{10}x_{47} - 4x_{47}^2 + x_{48}$$

$$\mathcal{R}_{34} = -4x_{42}^2 - 8x_{42}x_{48} - 4x_{48}^2 + x_{49}$$

$$\mathcal{R}_{35} = -9 + 7x_8 - x_8^2 - 8x_8x_{42} - 4x_8x_{49} + 24x_{42} - 16x_{42}^2 - 16x_{42}x_{49} + 12x_{49} - 4x_{49}^2 + x_{50}$$

$$\mathcal{R}_{36} = -1 - 4x_{50} - 4x_{50}^2 + x_{51}$$

$$\mathcal{R}_{37} = -x_{10}^2 - 8x_{10}x_{50} - 4x_{10}x_{51} - 16x_{50}^2 - 16x_{50}x_{51} - 4x_{51}^2 + x_{52}$$

$$\mathcal{R}_{38} = -x_8 - x_8^2 - 4x_8x_{10} - 4x_8x_{52} - 4x_{10} - 4x_{10}^2 - 8x_{10}x_{52} - 4x_{52} - 4x_{52}^2 + x_{53}$$

$$\mathcal{R}_{39} = -10 - 12x_{53} - 4x_{53}^2 + x_{54}$$

$$\mathcal{R}_{40} = -19 + 4x_9 - 4x_9^2 - 8x_9x_{23} - 16x_9x_{53} - 8x_9x_{54} + 4x_{23} - 4x_{23}^2 - 16x_{23}x_{53} - 8x_{23}x_{54} - 16x_{53} - 24x_{53}^2 - 16x_{53}x_{54} + 6x_{54} - 4x_{54}^2 + x_{55}$$

$$\mathcal{R}_{41} = -10 + x_9 - 25x_9^2 - 40x_9x_{23} - 20x_9x_{55} - 16x_{23}^2 - 16x_{23}x_{55} - 12x_{48} - 4x_{48}^2 - 4x_{55}^2 + x_{56}$$

$$\mathcal{R}_{42} = 21 + 24x_{48} - 8x_{48}^2 - 16x_{48}x_{56} - 2x_{56} - 4x_{56}^2 + x_{57}$$

$$\mathcal{R}_{43} = -5 + 4x_8 + 8x_9 + x_{10} - x_{10}^2 - 4x_{10}x_{57} - 2x_{19} - 4x_{57}^2 + x_{58} + 12x_{59} - 4x_{59}^2$$

$$\mathcal{R}_{44} = -6x_8 - 16x_8^2 - 64x_8x_9 + 8x_8x_{10} + 16x_8x_{19} - 12x_9 - 64x_9^2 + 16x_9x_{10} + 32x_9x_{19} + 3x_{10} - x_{10}^2 - 4x_{10}x_{19} + 3x_{19} - 4x_{19}^2 + x_{59}$$

$$\mathcal{R}_{45} = -12 + 8x_8 + 16x_9 - 4x_{19} - 4x_{58}^2 + 24x_{59} - 8x_{59}^2 + x_{60}$$

$$\mathcal{R}_{46} = -1 - 4x_{60} - 4x_{60}^2 + x_{61}$$

$$\mathcal{R}_{47} = -x_2x_{13} + x_{62}$$

$$\mathcal{R}_{48} = x_{12} - x_{13} - 2x_{22}x_{62} - x_{63}$$

$$\mathcal{R}_{49} = x_{12} - x_{13} - 2x_{26}x_{62} - x_{64}$$

$$\mathcal{R}_{50} = x_{12} - x_{13} - 2x_{28}x_{62} - 6x_{62} - x_{65}$$

$$\mathcal{R}_{51} = x_{12} - x_{13} - 2x_{30}x_{62} - x_{66}$$

$$\mathcal{R}_{52} = x_{12} - x_{13} - 2x_{32}x_{62} + 2x_{62} - x_{67}$$

$$\mathcal{R}_{53} = x_{12} - x_{13} - 4x_{34}x_{62} - 2x_{35}x_{62} - x_{68}$$

$$\mathcal{R}_{54} = x_{12} - x_{13} - 2x_{38}x_{62} - x_{69}$$

$$\mathcal{R}_{55} = 2x_{28}x_{62} - 2x_{41}x_{62} - 6x_{62} + x_{65} - x_{70}$$

$$\mathcal{R}_{56} = -x_8x_9 + x_8x_{10} + x_9 + x_9x_{10} - x_{10} - x_{10}^2 + x_{71}$$

$$\mathcal{R}_{57} = 1 + 2x_{11} + x_{11}^2 - x_{71}^2 + x_{72}$$
$$\mathcal{R}_{58} = -x_{12}^2 + 2x_{12}x_{13} + 2x_{12}x_{74} - x_{13}^2 - 2x_{13}x_{74} - x_{72}^2 + x_{73} - x_{74}^2$$
$$\mathcal{R}_{59} = -4x_{60}x_{62} - 2x_{61}x_{62} + x_{74}$$
$$\mathcal{R}_{60} = -x_9x_{62} - x_{13} + x_{75}x_{76} + x_{77}$$
$$\mathcal{R}_{61} = -1 + 4x_2 + 4x_2x_8 + 8x_2x_9 - 2x_2x_{19} + x_{75}$$
$$\mathcal{R}_{62} = -x_8x_{208} - x_{12} + x_{76} + x_{208}$$
$$\mathcal{R}_{63} = x_9x_{208} - 2x_{19}x_{62} + 2x_{32}x_{62} + x_{67} + x_{78}$$
$$\mathcal{R}_{64} = -x_{12}x_{63} + x_{13}x_{63} + x_{79}$$
$$\mathcal{R}_{65} = -x_{64}x_{79} + x_{80}$$
$$\mathcal{R}_{66} = -x_{65}x_{80} + x_{81}$$
$$\mathcal{R}_{67} = -x_{66}x_{81} + x_{82}$$
$$\mathcal{R}_{68} = -x_{67}x_{82} + x_{83}$$
$$\mathcal{R}_{69} = -x_{68}x_{83} + x_{84}$$
$$\mathcal{R}_{70} = -x_{69}x_{84} + x_{85}$$
$$\mathcal{R}_{71} = -x_{70}x_{85} + x_{86}$$
$$\mathcal{R}_{72} = -x_{77}x_{86} + x_{87}$$
$$\mathcal{R}_{73} = -x_{78}x_{87} + x_{88}$$
$$\mathcal{R}_{74} = -x_{88}^2 + x_{89}$$
$$\mathcal{R}_{75} = -x_4 - 2x_4x_{16} + 2x_{16} + x_{90}$$
$$\mathcal{R}_{76} = -x_{14}x_{90} + x_{91}$$
$$\mathcal{R}_{77} = -2x_1^2 + 2x_1x_{12} + 2x_1x_{13} + 2x_1x_{91} + 2x_1x_{92} - x_{12}^2 - 2x_{12}x_{91} - x_{13}^2 - 2x_{13}x_{92} + x_{17}x_{94} - x_{73}x_{89} - x_{91}^2 - x_{92}^2$$
$$\mathcal{R}_{78} = -x_4x_{15} + x_{15} - x_{15}x_{90} + x_{92}$$
$$\mathcal{R}_{79} = -x_{95}x_{98} + x_{98} + x_{100}$$
$$\mathcal{R}_{80} = -x_3x_{121} + x_{96}x_{200} - x_{99} - x_{100}x_{121}$$
$$\mathcal{R}_{81} = x_{101}x_{200} - x_{102}x_{124} - x_{103}$$
$$\mathcal{R}_{82} = x_{104}x_{200} - x_{105}x_{124} - x_{106}$$
$$\mathcal{R}_{83} = x_{107}x_{200} - x_{108}x_{124} - x_{109}$$
$$\mathcal{R}_{84} = x_{110}x_{200} - x_{111}x_{124} - x_{112}$$
$$\mathcal{R}_{85} = x_{113}x_{201} - x_{115}x_{124} - x_{116}$$
$$\mathcal{R}_{86} = x_{117}x_{201} - x_{118}x_{124} - x_{119}$$
$$\mathcal{R}_{87} = -x_{17}x_{121} + x_{124}$$
$$\mathcal{R}_{88} = x_{120} - x_{122}x_{124} - x_{123}$$
$$\mathcal{R}_{89} = -x_{124}x_{126} + x_{125} - x_{127}$$
$$\mathcal{R}_{90} = -x_{124}x_{129} + x_{128}x_{200} - x_{130}$$
$$\mathcal{R}_{91} = x_{131} - 2x_{131}x_{133} + x_{134}$$
$$\mathcal{R}_{92} = -x_{131}x_{134} - x_{134} + x_{135}$$
$$\mathcal{R}_{93} = -1 + x_{132}^2 + x_{134}^2 - x_{135}^2$$
$$\mathcal{R}_{94} = 1 - x_{132}x_{137} - x_{136} + x_{138}$$
$$\mathcal{R}_{95} = -1 + 2x_{136} - x_{136}^2 + 2x_{136}x_{138} + x_{137}^2$$
$$\mathcal{R}_{96} = 1 + x_{93}^2 - 2x_{93}x_{132} + x_{139}$$
$$\mathcal{R}_{97} = x_2 - x_{93}x_{182} + x_{139}x_{140} - x_{181}$$
$$\mathcal{R}_{98} = 1 + x_3^2 - 2x_3x_{132} + x_{141}$$
$$\mathcal{R}_{99} = -x_3x_{173} + x_{17} + x_{141}x_{142} - x_{172}$$
$$\mathcal{R}_{100} = -1 + 2x_{132}x_{184} - x_{143} - x_{184}^2$$
$$\mathcal{R}_{101} = -1 + x_{95} + x_{143}x_{144} - x_{172} - x_{173}x_{184}$$
$$\mathcal{R}_{102} = -x_{132}x_{146} - x_{145} + x_{147}$$
$$\mathcal{R}_{103} = -1 - x_{145}^2 + 2x_{145}x_{147} + x_{146}^2$$
$$\mathcal{R}_{104} = 1 - x_{132}x_{149} - x_{148} + x_{150}$$
$$\mathcal{R}_{105} = -1 + 2x_{148} - x_{148}^2 + 2x_{148}x_{150} + x_{149}^2$$

$$\mathcal{R}_{106} = -x_{132}x_{152} - x_{151} + x_{153}$$
$$\mathcal{R}_{107} = -1 - x_{151}^2 + 2x_{151}x_{153} + x_{152}^2$$
$$\mathcal{R}_{108} = -x_{132}x_{155} - x_{154} + x_{156}$$
$$\mathcal{R}_{109} = -1 - x_{154}^2 + 2x_{154}x_{156} + x_{155}^2$$
$$\mathcal{R}_{110} = -x_{132}x_{158} - x_{157} + x_{159}$$
$$\mathcal{R}_{111} = -1 - x_{157}^2 + 2x_{157}x_{159} + x_{158}^2$$
$$\mathcal{R}_{112} = -x_{132}x_{161} - x_{160} + x_{162}$$
$$\mathcal{R}_{113} = -1 - x_{160}^2 + 2x_{160}x_{162} + x_{161}^2$$
$$\mathcal{R}_{114} = 1 - x_{132}x_{164} - x_{163} + x_{165}$$
$$\mathcal{R}_{115} = -1 + 2x_{163} - x_{163}^2 + 2x_{163}x_{165} + x_{164}^2$$
$$\mathcal{R}_{116} = -x_{132}x_{167} - x_{166} + x_{168}$$
$$\mathcal{R}_{117} = -1 - x_{166}^2 + 2x_{166}x_{168} + x_{167}^2$$
$$\mathcal{R}_{118} = -x_{132}x_{170} - x_{169} + x_{171}$$
$$\mathcal{R}_{119} = -1 - x_{169}^2 + 2x_{169}x_{171} + x_{170}^2$$
$$\mathcal{R}_{120} = 1 - x_{132}x_{173} - x_{172} + x_{174}$$
$$\mathcal{R}_{121} = -1 + 2x_{172} - x_{172}^2 + 2x_{172}x_{174} + x_{173}^2$$
$$\mathcal{R}_{122} = -x_{132}x_{176} - x_{175} + x_{177}$$
$$\mathcal{R}_{123} = -1 - x_{175}^2 + 2x_{175}x_{177} + x_{176}^2$$
$$\mathcal{R}_{124} = -x_{132}x_{179} - x_{178} + x_{180}$$
$$\mathcal{R}_{125} = -1 - x_{178}^2 + 2x_{178}x_{180} + x_{179}^2$$
$$\mathcal{R}_{126} = 1 - x_{132}x_{182} - x_{181} + x_{183}$$
$$\mathcal{R}_{127} = -1 + 2x_{181} - x_{181}^2 + 2x_{181}x_{183} + x_{182}^2$$
$$\begin{aligned}\mathcal{R}_{128} = &-5 - 2x_0 - 3x_0^2 - 2x_0x_2 - x_1^2 - 2x_2^2 - x_3^2 - x_4^2 + 2x_8 \\ &- x_8^2 - x_9^2 - x_{10}^2 - x_{11}^2 - x_{12}^2 - x_{13}^2 - x_{14}^2 \\ &- x_{15}^2 - 4x_{16}^2 - x_{17}^2 - 2x_{93}^2 + 2x_{95} - 2x_{95}^2 - x_{96}^2 \\ &- x_{97}^2 - x_{101}^2 - x_{104}^2 - x_{107}^2 - x_{110}^2 - x_{113}^2 \\ &- x_{114}^2 - x_{117}^2 - x_{120}^2 - x_{121}^2 - x_{125}^2 - x_{128}^2 \\ &+ x_{131} - x_{184}^2\end{aligned}$$
$$\begin{aligned}\mathcal{R}_{129} = &\ 2x_3 - 4x_3x_8 - 4x_3x_9 - 4x_3x_{10} - 4x_3x_{11} - 4x_3x_{12} - 4x_3x_{13} - 4x_3x_{14} \\ &- 4x_3x_{15} - 8x_3x_{16} + x_{184}\end{aligned}$$
$$\mathcal{R}_{130} = -464 + x_0 - x_{185}^2$$
$$\mathcal{R}_{131} = -x_{132}x_{187} + x_{196}$$
$$\mathcal{R}_{132} = -1 + x_{186}^2 + x_{187}^2 - x_{196}^2$$
$$\mathcal{R}_{133} = -x_{189}x_{190} + x_{197}$$
$$\mathcal{R}_{134} = -1 + x_{188}^2 + x_{189}^2 - x_{197}^2$$
$$\mathcal{R}_{135} = -x_{179}^2 + x_{198}$$
$$\mathcal{R}_{136} = x_{187} - 2x_{192}x_{198} - x_{198}$$
$$\mathcal{R}_{137} = -1 - 4x_{187} - 8x_{187}x_{191} + x_{190}$$
$$\mathcal{R}_{138} = -x_{132} - x_{186}x_{193} + x_{190}$$
$$\mathcal{R}_{139} = -x_{180} - x_{186}x_{194} + x_{188}$$
$$\mathcal{R}_{140} = -2x_{16} - 4x_{179}x_{195} + x_{189}$$

$$
\begin{aligned}
\mathcal{R}_{141} &= 9 + 2x_0^2 + 2x_0x_2 - 2x_0x_{137} - 2x_0x_{146} + x_1^2 - 2x_1x_{173} + x_2^2 \\
&\quad - 2x_2x_{146} + x_4^2 - 2x_4x_{164} - 2x_8 + x_8^2 - 2x_8x_{149} + x_9^2 - 2x_9x_{152} \\
&\quad + x_{10}^2 - 2x_{10}x_{155} + x_{11}^2 - 2x_{11}x_{158} + x_{12}^2 - 2x_{12}x_{161} \\
&\quad + x_{13}^2 - 2x_{13}x_{167} + x_{14}^2 - 2x_{14}x_{170} + x_{15}^2 - 2x_{15}x_{176} \\
&\quad - 4x_{16}x_{179} + x_{93}^2 - 2x_{93}x_{182} - x_{132}x_{199} + x_{138}^2 - 2x_{147} \\
&\quad + x_{147}^2 + 2x_{149} + x_{150}^2 - 2x_{153} + x_{153}^2 - 2x_{156} + x_{156}^2 \\
&\quad - 2x_{159} + x_{159}^2 - 2x_{162} + x_{162}^2 + x_{165}^2 - 2x_{168} + x_{168}^2 \\
&\quad - 2x_{171} + x_{171}^2 + x_{174}^2 - 2x_{177} + x_{177}^2 + 2x_{179}^2 - 2x_{180} \\
&\quad + x_{180}^2 + x_{183}^2 + x_{199} - x_{202}^2 - x_{203}^2 - x_{204} \\
&\quad - x_{204}^2 \\
\mathcal{R}_{142} &= -x_{97}x_{200} + x_{121} \\
\mathcal{R}_{143} &= -x_{114}x_{201} + x_{121} \\
\mathcal{R}_{144} &= -1 - 4x_{16}^2 - x_{137}^2 - x_{146}^2 - x_{149}^2 - x_{152}^2 - x_{155}^2 \\
&\quad - x_{158}^2 - x_{161}^2 - x_{164}^2 - x_{167}^2 - x_{170}^2 - x_{173}^2 \\
&\quad - x_{176}^2 + x_{179}^2 - x_{182}^2 - x_{202}^2 - x_{203}^2 - x_{204} \\
&\quad - x_{204}^2 \\
\mathcal{R}_{145} &= -1 - 4x_{99}^2 - 4x_{103}^2 - 4x_{106}^2 - 4x_{109}^2 - 4x_{112}^2 - 4x_{116}^2 \\
&\quad - 4x_{119}^2 + x_{121}^2 - 4x_{123}^2 - 4x_{127}^2 - 4x_{130}^2 - x_{205}^2 \\
&\quad - x_{206}^2 - x_{207} - x_{207}^2 \\
\mathcal{R}_{146} &= -x_2x_{12} + x_{208} \\
\mathcal{R}_{147} &= x_{95}x_{210} - 2x_{96}x_{146} - 2x_{101}x_{149} - 2x_{104}x_{152} - 2x_{107}x_{155} - 2x_{110}x_{158} \\
&\quad - 2x_{113}x_{161} - 2x_{117}x_{167} - 2x_{120}x_{170} - 2x_{125}x_{176} - 2x_{128}x_{179} - 2x_{132}x_{210} \\
&\quad + 2x_{146}x_{147} + 2x_{149} + 2x_{149}x_{150} + 2x_{152}x_{153} + 2x_{155}x_{156} + 2x_{158}x_{159} \\
&\quad + 2x_{161}x_{162} + 2x_{167}x_{168} + 2x_{170}x_{171} + 2x_{176}x_{177} + 2x_{179}x_{180} + x_{209} \\
\mathcal{R}_{148} &= -10 - x_{95}x_{209} + x_{96}^2 - 2x_{96}x_{145} + x_{101}^2 - 2x_{101}x_{148} + x_{104}^2 \\
&\quad - 2x_{104}x_{151} + x_{107}^2 - 2x_{107}x_{154} + x_{110}^2 - 2x_{110}x_{157} + x_{113}^2 \\
&\quad - 2x_{113}x_{160} + x_{117}^2 - 2x_{117}x_{166} + x_{120}^2 - 2x_{120}x_{169} + x_{125}^2 \\
&\quad - 2x_{125}x_{175} + x_{128}^2 - 2x_{128}x_{178} + 2x_{145}x_{147} + 2x_{148} + 2x_{148}x_{150} \\
&\quad + 2x_{151}x_{153} + 2x_{154}x_{156} + 2x_{157}x_{159} + 2x_{160}x_{162} + 2x_{166}x_{168} + 2x_{169}x_{171} \\
&\quad + 2x_{175}x_{177} + 2x_{178}x_{180} + x_{210} \\
\mathcal{R}_{149} &= x_{95}x_{212} - 2x_{97}x_{137} - 2x_{114}x_{164} - 2x_{121}x_{173} - 2x_{132}x_{212} + 2x_{137} \\
&\quad + 2x_{137}x_{138} + 2x_{164} + 2x_{164}x_{165} + 2x_{173} + 2x_{173}x_{174} + x_{211} - x_{212} \\
\mathcal{R}_{150} &= -2 - x_{95}x_{211} + x_{97}^2 - 2x_{97}x_{136} + x_{114}^2 - 2x_{114}x_{163} + x_{121}^2 \\
&\quad - 2x_{121}x_{172} + 2x_{136} + 2x_{136}x_{138} + 2x_{163} + 2x_{163}x_{165} + x_{172}^2 \\
&\quad - x_{173}^2 + x_{211} + x_{212} \\
\mathcal{R}_{151} &= -1 - x_2 - 2x_2x_{213} + x_4 \\
\mathcal{R}_{152} &= -464x_0^2 + x_{93}
\end{aligned}
$$